

B.Tech. DEGREE EXAMINATION, NOVEMBER 2019
Third Semester

18MES201T – ENGINEERING MECHANICS

(For the candidates admitted during the academic year 2018-2019 onwards)
(use of non programmable calculators is allowed)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

- The resultant of concurrent forces can be determined by
 (A) Parallelogram law (B) Polygon law
 (C) Lamis theorem (D) Triangle law of forces
- The conditions of equilibrium for coplanar non-concurrent forces are
 (A) $\sum H = 0, \sum V = 0$ (B) $\sum H = 0, \sum M = 0$
 (C) $\sum V = 0, \sum M = 0$ (D) $\sum H = 0, \sum V = 0, \sum M = 0$
- The rotational tendency of a force is called
 (A) Shear force (B) Moment
 (C) Couple (D) Centroid
- Two concurrent forces acts at an angle of 30° . The resultant force is 15 N and one of the force is having magnitude of 10 N. What is the magnitude of the force
 (A) 5.48 N (B) 12.6 N
 (C) 19.2 N (D) 2.8 N
- The coefficient of static friction is always _____ that of kinetic friction.
 (A) Less than (B) Equal to
 (C) Greater than (D) Independent of
- The efficiency of a screw jack is given by
 (A) $\frac{\tan(\alpha + \phi)}{\tan \alpha}$ (B) $\frac{\tan \alpha}{\tan(\alpha + \phi)}$
 (C) $\frac{\tan(\alpha - \phi)}{\tan \alpha}$ (D) $\frac{\tan \alpha}{\tan(\alpha - \phi)}$
- The efficiency of a screw jack is maximum when
 (A) $\alpha = 45^\circ + \frac{\phi}{2}$ (B) $\alpha = 45^\circ - \frac{\phi}{2}$
 (C) $\alpha = 90^\circ + \phi$ (D) $\alpha = 90^\circ - \phi$

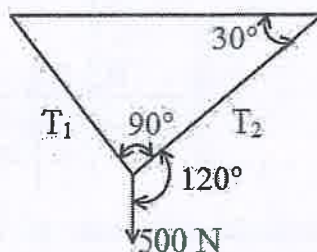
8. The condition for self locking of a screw is
 (A) $\alpha = \phi$ (B) $\alpha < \phi$
 (C) $\alpha > \phi$ (D) $\alpha \leq \phi$
9. The area moment of inertia of a section about XX axis is given by
 (A) $I_{xx} = \int x^2 dA$ (B) $I_{xx} = \int x^2 dV$
 (C) $I_{xx} = \int y^2 dA$ (D) $I_{xx} = \int y^2 dV$
10. The moment of inertia of triangular area about its base is given by a relation
 (A) $I = \frac{bh}{12}$ (B) $I = \frac{bh^3}{12}$
 (C) $I = \frac{bh^2}{36}$ (D) $I = \frac{bh^3}{36}$
11. The mass moment of inertia of a circular disc of radius 'r' about its radial axis is
 (A) $\frac{mr^2}{2}$ (B) $\frac{mr^2}{4}$
 (C) $\frac{mr^2}{8}$ (D) mr^2
12. The relation between the centroid of an area and the volume generated by that area is given by
 (A) Parallel axis theorem (B) Perpendicular axis theorem
 (C) Pappus-guldinus theorem (D) Pappus-guldinus theorem-II
13. Two masses 2 kg and 8 kg are moving with a equal kinetic energy. The ratio of magnitudes of thick momentum is
 (A) 0.25 (B) 0.50
 (C) 0.625 (D) 1.00
14. During inelastic collision of two particles which one of the following is conserved
 (A) Total linear momentum only (B) Total kinetic energy only
 (C) Both linear momentum and kinetic energy (D) Neither linear momentum nor kinetic energy
15. A particle is projected vertically upward with an initial velocity 'U'. If g is the acceleration due to gravity, then which one of the following is the greatest height 'h' attained by it?
 (A) $h = \frac{2u^2}{g}$ (B) $h = \frac{u^2}{g}$
 (C) $h = \frac{1}{2} \frac{u^2}{g}$ (D) $h = \frac{1}{4} \frac{u^2}{g}$
16. The coefficient of restitution of a perfectly plastic impact is
 (A) 0 (B) 1
 (C) 2 (D) ∞

17. Instantaneous centre of a body rolling without sliding on a stationary curved surface lies
 (A) At the point of contact (B) On the common normal at the point of contact
 (C) On the common tangent at the point of contact (D) At the centre of curvature of the stationary surface
18. A circular object of radius 'r' rolls without slipping on a horizontal level floor with the center having velocity 'V'. The velocity at the point of contact between the object and the floor is
 (A) Zero (B) V in the direction of motion
 (C) V opposite to the direction of motion (D) V vertically upward from the floor
19. A train of weight 5000 kN is pulled on a level track with a constant speed of 10 m/s. The frictional force is 75 kN. The power of the engine is
 (A) 375 kW (B) 750 kW
 (C) 1875 kW (D) 3000 kW
20. A circular disc rolls down without slip on an incline plane. The ratio of its rotational kinetic energy to the total energy is
 (A) 1/4 (B) 1/2
 (C) 1/3 (D) 2/3

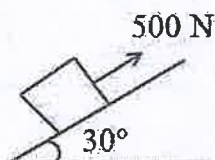
PART – B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. A weight of 500 N is supported by two metallic ropes as shown below. Find the values of tension T_1 and T_2 .



22. If the maximum and minimum resultant forces of the two forces acting on a particle are 40 kN and 10 kN respectively. Then determine the two forces acting on the particle.
23. The block shown below is kept in equilibrium and prevented from sliding down by applying a force of 500 N. The coefficient of friction is $\frac{\sqrt{3}}{5}$. Determine the weight of the block.



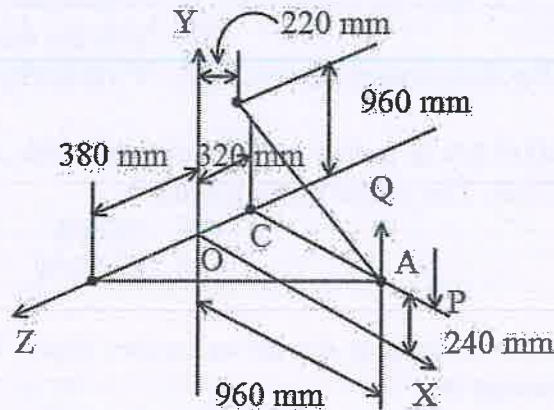
24. A load of 2 kN is to be lifted by a screw jack having threads of 10 mm pitch. The efficiency of the jack at this load is 50%. Determine the effort applied at the end of a handle of 500 mm length.
25. State and prove parallel axis theorem.
26. A mass m_1 of 100 kg travelling with a uniform velocity of 5 m/s along a line collides with a stationary mass m_2 of 1000 kg. After collision both the masses travel together with same velocity. Determine the coefficient of restitution.

27. A car moving with uniform acceleration covers 450 m in a 5 second interval and covers 700 m in the next 5 second interval. The acceleration of the car is?

PART – C ($5 \times 12 = 60$ Marks)

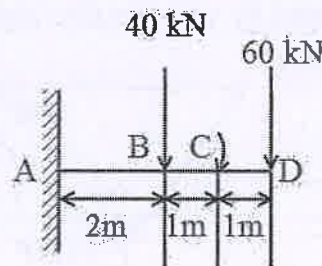
Answer ALL Questions

28. a. Three cables are connected at A, where the forces P and Q are applied as shown below. Knowing that $Q=0$. Find the value of P for which the tension in cable AD is 305 N.

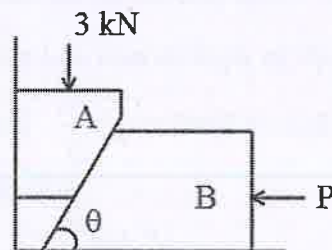


(OR)

- b. Two vertical forces and a couple of magnitude 20 kNm, acting on a horizontal rod, which is fixed at A.
- Determine the resultant of the system
 - Reduce the force couple system at A



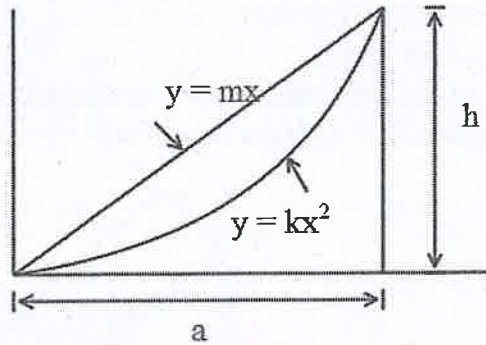
29. a. Block A supports a pipe column and rests as shown on wedge B. knowing that the coefficient of static friction at all surfaces of contact is 0.25 and that $\theta = 45^\circ$. Determine the smallest force 'P' required to raise block A.



(OR)

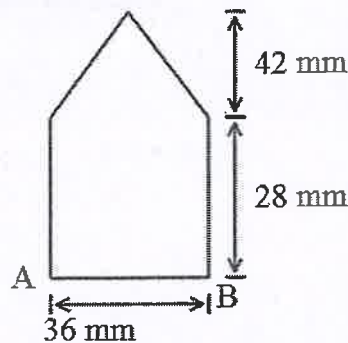
- b. A screw jack has a square thread of 75 mm mean diameter and 15 mm pitch. The load on the jack is 6 kN. The coefficient of friction at the screw thread is 0.05.
- Find the tangential force to be applied to the jack at 360 mm radius to raise the load
 - State whether the jack is self locking
- If is so find, the torque required to lower the load. If not, find the torque applied to kept the load from descending.

30. a. Determine by direct integration the centroid of the area shown. Express your answer in terms of 'a' and 'h'.



(OR)

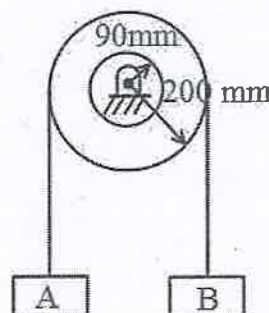
- b. Determine the moment of inertia I_x and I_y of the area shown with respect to centroidal axes respectively parallel and perpendicular to side AB.



31. a. The pilot of an aeroplane flying horizontally at a height of 3000 m with a constant speed of 510 km/hr aims to hit a target on the ground. At what distance from the target should he release the bomb in order to hit the target? At what angle would the target appear to him from that distance?

(OR)

- b. A bullet weighs 0.7 N and is moving with a velocity of 275 m/s hits centrally a 40 N block of wood moving away at 16 m/s and gets embedded in it find the velocity of the bullet after the impact and the amount of kinetic energy lost.
32. a. Two blocks and a pulley are connected by inextensible cords as shown. The pulley has an initial angular velocity of 0.8 rad/s in counter clockwise direction and a constant angular acceleration of 1.8 rad/s² in clockwise direction after 5 sec of motion, determine the velocity and position of the block A and B.



(OR)

- b. A bar AB of length 1.2 m slides in xy plane as shown below. The velocity of the point A is 5 m/s towards right. Determine
- The angular velocity of the bar
 - The velocity of the end B
 - The velocity of the mid point of the bar at the instant when the axis of the bar makes an angle of 30° with the horizontal

